

- (b) Two isolated objects, X and Y travel along the same straight line with speeds 3.5 m s^{-1} and 2.0 m s^{-1} respectively as shown in Fig. 1.1. The objects collide elastically and continue to travel along the same straight line after the collision.



Fig. 1.1

Object X has a mass of 0.5 kg and object Y has a mass of 0.25 kg .

The variation with time of the force exerted by object X on object Y during the collision is shown in Fig. 1.2.

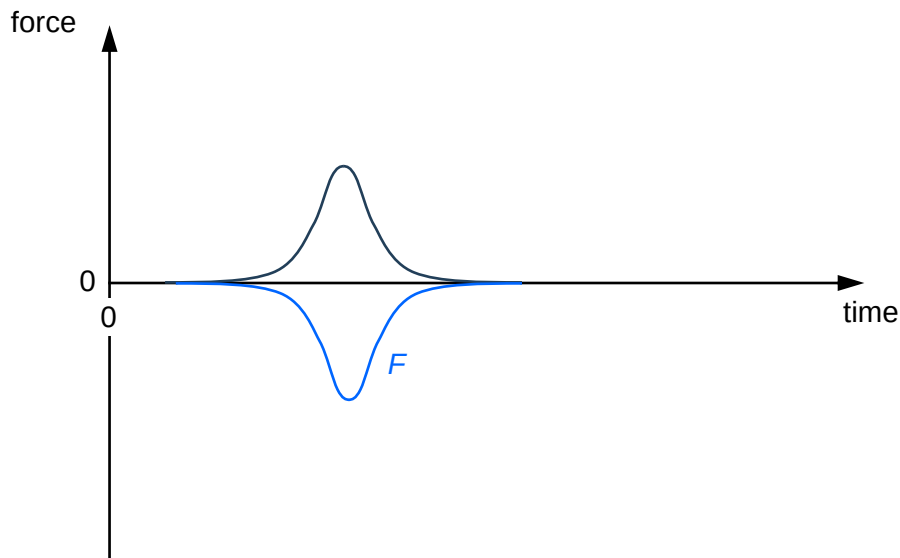


Fig. 1.2

- (i) Sketch on Fig. 1.2 the variation with time of the force exerted by object Y on object X during the collision. Label this line F . [1]
- (ii) State and explain whether, during the collision it is possible for both objects to be at rest simultaneously.

No is it not possible. This is because the total initial momentum of both objects is non zero, hence by the principle of conservation of linear momentum, there will not be any instance where both objects can be at rest at the same time where total momentum is zero. [2]

- (iii) The area under the graph given in Fig. 1.2 is 0.50 N s. Use the information in (b) to calculate the velocity of object Y after the collision.

The area under the graph = impulse on Y

The change in momentum in Y is then 0.50 Ns.

$$\begin{aligned} m_y v_y - m_y u_y &= -0.50 & \text{B1} \\ (0.25)v_y - (0.25)(2.0) &= 0.50 & \text{M1} \\ v_y &= 4.0 \text{ m s}^{-1} & \text{A1} \end{aligned}$$

velocity of object Y = m s⁻¹ [3]

[Total: 9]

- 2 (a) A student performs an experiment to determine the specific latent heat of fusion of ice. The student has two sets of apparatus next to each other on the laboratory bench, as shown in Fig. 2.1 and Fig. 2.2.

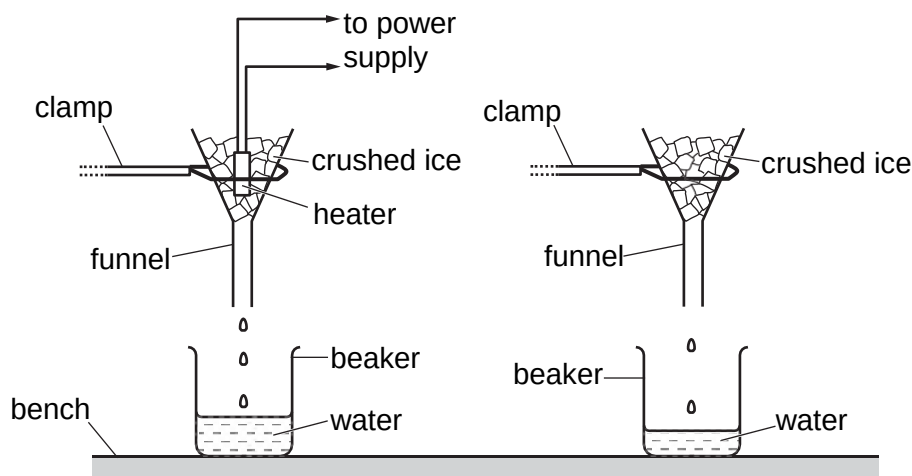


Fig. 2.1

Fig. 2.2

Both funnels are identical and have the same mass of crushed ice at 0°C .

The current in the heater is 5.0 A and the potential difference across it is 12 V .

Fig. 2.3 shows the variation of mass of water m collected in each beaker with time t .

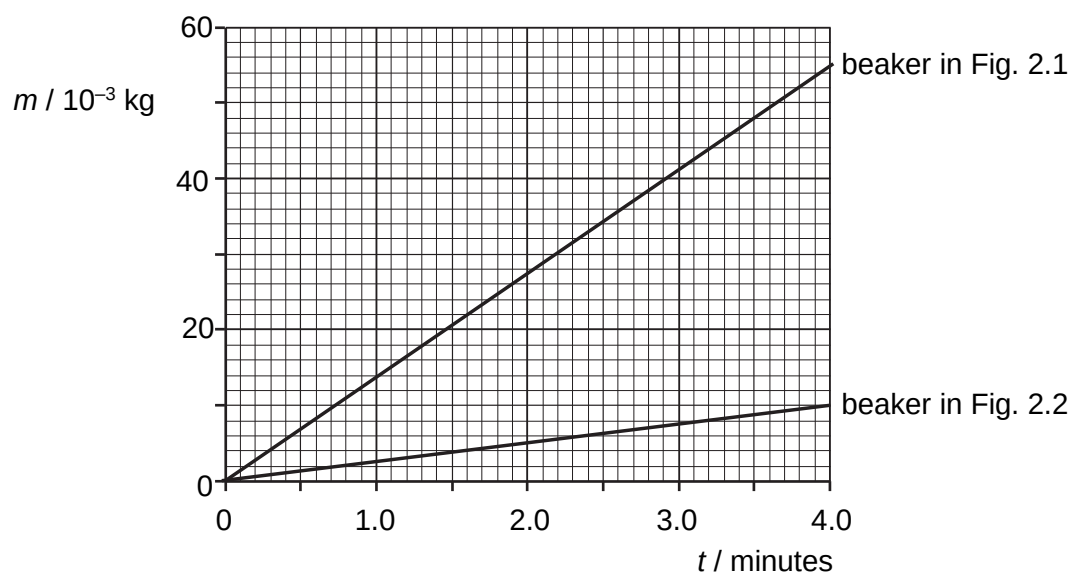


Fig. 2.3

- (i) Explain why the gradients of the two graphs are different.

Rate of mass of water collected OR rate of mass of ice melted in Fig. 2.1 is greater due
to heat from heater and surroundings compared to Fig. 2.2 from surroundings only [B1] [1]

- (ii) Use Fig. 2.3 to show that the specific latent heat of fusion of ice is about $3 \times 10^5 \text{ J kg}^{-1}$.

$$\text{rate of ice melted} = \frac{m}{t} = \text{gradient}$$

$$\begin{aligned} \text{rate of ice melted by heater} &= \left(\frac{m}{t} \right)_{\text{heater+surr}} - \left(\frac{m}{t} \right)_{\text{surr}} \\ &= \frac{55 \times 10^{-3}}{4.0 \times 60} - \frac{10 \times 10^{-3}}{4.0 \times 60} \quad [\text{M1}] \\ &= 1.875 \times 10^{-4} \text{ kg s}^{-1} \end{aligned}$$

$$Pt = ml_f$$

$$IV = \left(\frac{m}{t} \right) l_f$$

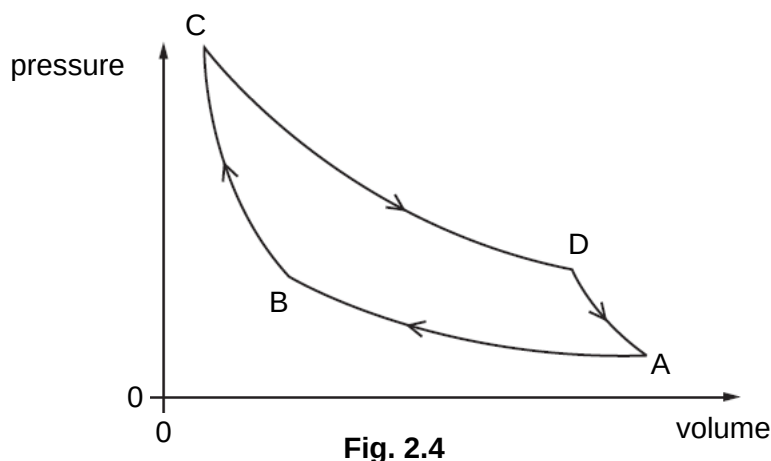
$$\begin{aligned} l_f &= \frac{(5.0)(12)}{1.875 \times 10^{-4}} \quad [\text{M1}] \\ &= 3.2 \times 10^5 \text{ J kg}^{-1} \\ &= 3 \times 10^5 \text{ J kg}^{-1} \quad [\text{A0}] \end{aligned}$$

$$\text{specific latent heat of fusion} = \dots\dots\dots 3 \times 10^5 \text{ J kg}^{-1} \quad [2]$$

- (b) A heat engine, such as a car engine, is a device that converts thermal energy into mechanical work.

When the heat engine operates, a fixed amount of gas expands and contracts repeatedly in a cylinder with a piston.

The cycle of expansion and contraction for a fixed quantity of an ideal gas is illustrated graphically in Fig. 2.4.



There are four stages in the cycle.

stage	description
A to B	a slow compression of the gas at constant temperature
B to C	a sudden compression of the gas causing an increase in temperature
C to D	a slow expansion of the gas at constant temperature
D to A	a sudden expansion back to its original pressure, volume, and temperature

(i) Explain each of the following facts about the cycle:

- 1 During stage B to C, the piston causes a sudden compression of the gas, causing an increase in temperature, with reference to the kinetic theory of gases.

Speed / kinetic energy of gas molecules increase when hit by the moving piston,
so temperature of gas increases [B1][1]

- 2 At the end of all four stages, the change in internal energy of the gas is zero.

The gas returned to its original temperature so no change in internal energy [B1]
.....[1]

(ii) Complete the table in Fig. 2.5 for the cycle.

stage	thermal energy supplied to gas / J	work done on gas / J	increase in internal energy of gas / J
A to B	-702	702	0
B to C	0	844	844
C to D	936	-936	0
D to A	0	-844	-844

[B1] for both

Fig. 2.5

[1]

(iii) Determine the efficiency of the heat engine.

Show your working clearly.

$$\eta = \frac{\text{net work done BY gas}}{\text{thermal energy supplied}} \times 100\%$$

$$= \frac{(936 + 844) - (702 + 844)}{936} \times 100\% \quad [\text{M1}]$$

$$= \frac{234}{936} \times 100\%$$

$$= 25\% \quad [\text{A1}]$$

$$\text{efficiency} = \frac{25}{100} \times 100\% = 25\% \quad [\text{2}]$$

[Total: 8]